

INTRODUCTION TO BUSINESS ANALYTICS

MEDICAL INSURANCE ANALYSIS REPORT

Submitted By,

Ashwin Balaji Manjula Paramashivam (101539897)

Beula Evangelin Nicolas Suresh (101545182)

Mahima Srinivasan(101546056)

Purusothaman Loganathan (101553143)

Introduction:

The insurance industry collects large volumes of data, which can be used to predict and understand the behavior of insurance charges for individuals. This report focuses on analyzing an insurance dataset using Python for exploratory data analysis (EDA) and linear regression modeling. The dataset includes key variables like age, sex, BMI, number of children, smoking status, and region, along with the corresponding insurance charges. Through EDA, patterns, relationships, and outliers within the dataset are explored to provide insights into the data distribution and potential predictors for charges. Following the analysis, linear regression is applied to model the relationship between these features and insurance charges. The goal is to better understand the factors influencing insurance costs and to create a predictive model that can estimate charges based on an individual's attributes.

This approach provides a comprehensive understanding of the dataset and the potential to leverage data-driven models to predict and explain insurance charges efficiently.

1. Exploratory Data Analysis (EDA)

1.1 Data Cleaning

It was vital to clean the dataset before beginning the analysis. The following were the outcomes.

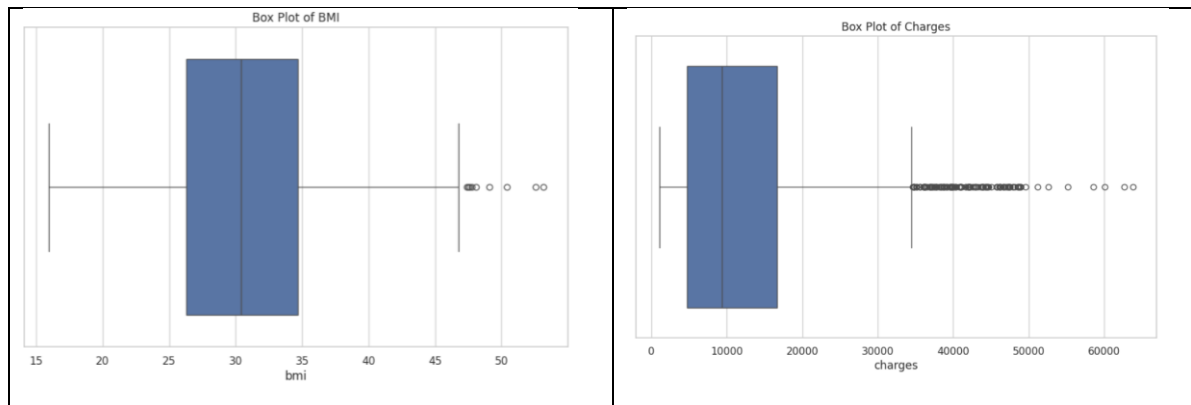
- **Checking for Missing Data:** Ensuring there are no null values in critical columns.

```
age      0
sex      0
bmi      0
children 0
smoker   0
region   0
charges  0
dtype: int64
```

- **Duplicates:** There was one duplicate record in the database. It was removed using “drop duplicates”. Earlier there were 1338 rows, after removal, there are 1337.
- **Statistical Summary:** Understanding the distribution and summary statistics of numerical columns (like age, BMI, charges).

- **Handling Outliers:** We used the z-score method to look for outliers. We took care of the extreme values that we found in the BMI and charges columns to avoid skewing the results.

Outliers visualized for BMI and Charges Features



Handling Outliers using Z-score

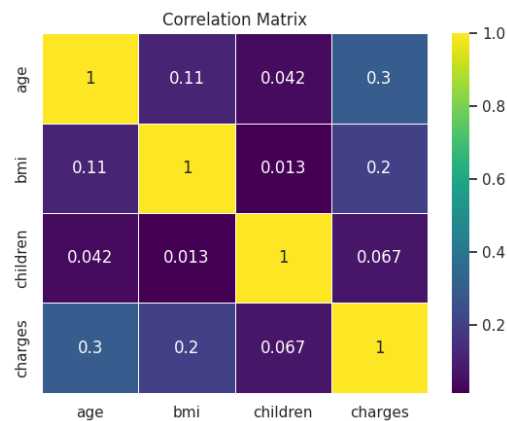
	age	sex	bmi	children	smoker	region	charges
0	19	female	27.900	0	yes	southwest	16884.92400
1	18	male	33.770	1	no	southeast	1725.55230
2	28	male	33.000	3	no	southeast	4449.46200
3	33	male	22.705	0	no	northwest	21984.47061
4	32	male	28.880	0	no	northwest	3866.85520

1.2 Correlation Matrix

Checking for correlations between the features and target variable (charges). Features like BMI, age, and smoking status are expected to have strong relationships with insurance charges.

Moderate Correlations: Age and BMI have a moderate positive correlation with charges with 0.3 and 0.2 correlation. As age or BMI increases, the insurance charges tend to increase as well.

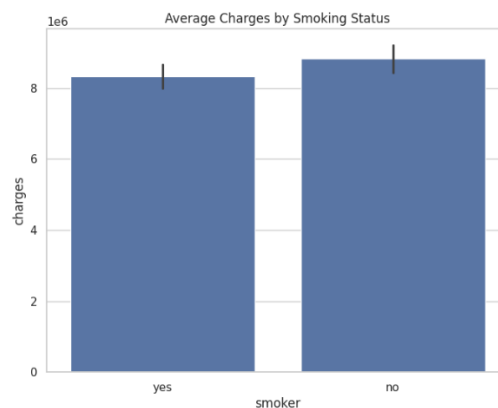
Children: The number of children has minimal impact on charges with the least of 0.067, suggesting that it's not a significant factor for this dataset.



1.3 The influence of Smoking status:

An additional intriguing discovery is the effect of smoking on medical expenses. Costs are typically far higher for smokers than for non-smokers. On average:

- Smokers frequently incur expenses between \$30,000 and \$40,000.
- Conversely, non-smokers typically pay between \$10,000 and \$15,000.

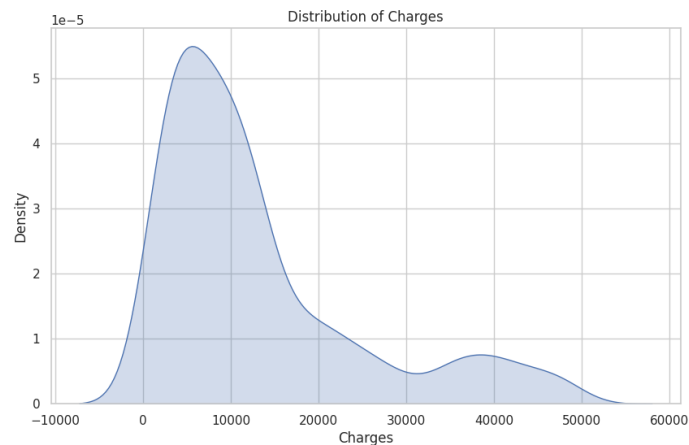


It further indicates that smoking increases healthcare costs significantly, which may have consequences for healthcare policies that attempt to lower smoking rates.

2. Data Visualization and Insights

2.1 Distribution of Charges

To understand how the insurance charges are distributed across the dataset using KDE. It is a non-parametric method for calculating a continuous variable's probability density function.

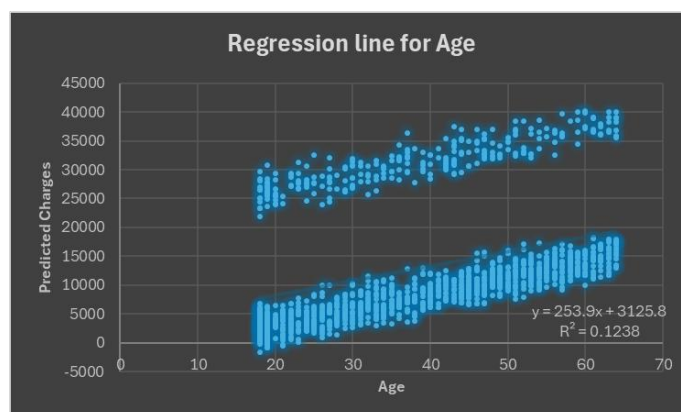


- With the majority of the charges clustered between 0 and 15,000, the plot displays a right-skewed distribution. A few number of the high charges surpass 30,000.
- The majority of people pay less for insurance; a small number pay much more.

Interpreting the Plot

- **Right Skewed:** The majority of data is concentrated on the left side (lower charges).
- **Peak:** The highest density (most common charges) occurs around 10,000.
- **Long Tail:** Fewer instances of higher charges above 30,000.
- **Insights:** This suggests that the majority of individuals in the dataset incurred lower charges, while a small portion faced very high charges.
- This indicates that most policyholders are either healthy or have low-risk factors, while a minority, likely those with more risk factors such as smoking, high BMI, or age, are driving the higher charges.

2.2 Age vs Predicted Charges



The scatter plot illustrates the relationship between age and predicted charges

$$y=253.9x+3125.8$$

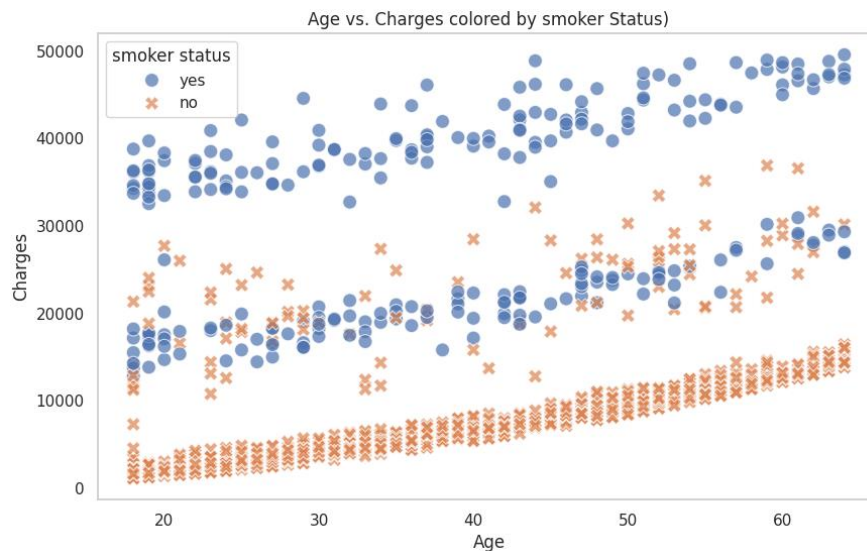
This equation shows that predicted charges rise by **253.9 units** for every additional year of age, starting with a baseline of **3125.8 units**. The **R² value of 0.1238** indicates that age explains only **12.38%** of the variation in predicted charges, suggesting a weak correlation. The plot reveals two distinct clusters, likely indicating different groups with varying cost structures, possibly smokers and non-smokers. While healthcare costs do tend to increase slightly with age, the low R² and the distinct clusters suggest that factors like smoking status or chronic health conditions play a larger role in determining charges. Age by itself is not a strong predictor of healthcare costs

2.3 Age vs Charges Coloured by Smoker Status

Goal: The scatter plot depicts the relationship between **age** (x-axis) and **charges** (y-axis) thus showing how age and insurance costs relate to one another while accounting for smoker status.

Each point represents an individual from the insurance dataset.

- **Blue circles:** Represent smokers.
- **Orange crosses:** Represent non-smokers.



Key Technical Insights

A. Overall Trend

- **Smokers have higher charges:** Smokers (blue circles) generally incur significantly higher healthcare costs compared to non-smokers (orange crosses).
- **Charges increase with age:** As individuals get older, their insurance charges tend to increase, especially for smokers.

B. Smoker vs. Non-Smoker Comparison

- **Non-smokers:** Have lower and more concentrated charges, mostly below **\$20,000**, reflecting lower healthcare costs.
- **Smokers:** Show a wide range of charges, with many exceeding **\$30,000**, indicating greater variability in healthcare costs, likely due to smoking-related health issues.

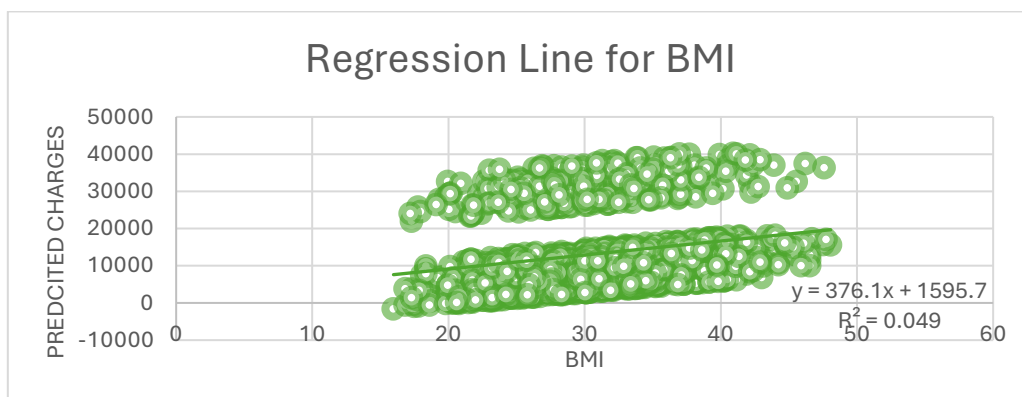
C. Age-Based Patterns

- **Young smokers already have high charges:** Even younger smokers in their 20s and 30s incur higher charges than non-smokers of the same age.
- **Older smokers face even higher charges:** For individuals over 40, the cost gap widens significantly, with smokers' charges continuing to rise more sharply.

Observation: Across all age categories, smokers (blue) tend to be charged higher, whereas non-smokers (orange) tend to be charged lower.

Smokers incur significantly higher medical charges than non-smokers, with charges increasing sharply as they age. Non-smokers have more consistent, lower charges, while smokers show a wide range of higher costs, especially in older age groups. Age has a stronger impact on increasing costs for smokers than for non-smokers.

2.3 BMI vs. Predicted Charges:



The scatter plot illustrates the relationship between BMI and Predicted Charges represented by the regression equation $y=376.1x+1595.7$. This equation indicates that healthcare costs increase by **376.1 units** for every unit increase in BMI, starting from a baseline cost of **1595.7 units** at a BMI of zero. However, the **R² value of 0.049** reveals that BMI only accounts for **4.9%** of the variation in predicted charges, indicating it is a weak predictor. The plot displays two distinct clusters: one with predicted charges between **0 and 20,000** and another between **25,000 and 40,000**, suggesting that other factors such as smoking status, gender, or health conditions play a more significant role in determining healthcare charges. Although higher BMI is associated with increased health risks and costs, the data's spread suggests that variables like smoking or pre-existing conditions have a stronger influence. The low R² indicates that the linear regression model lacks fit, pointing to the need for additional factors to more accurately predict expenses, as the observed clusters likely reflect the impact of unaccounted variables, particularly smoking status.

3. Linear Regression Analysis

3.1 Model Building

We used model to predict charges that incurred. Here, we used Linear Regression Data Analysis method where BMI, Smoker, Age are independent variables and Actual Charges is Dependent variable. Smoking status was coded as a binary variable (1 for smokers and 0 for non-smokers).

Equation: The regression equation derived from the model is:

$$Y = \beta_0 + \beta_1 \cdot X_1 + \beta_2 \cdot X_2 + \beta_3 \cdot X_3$$

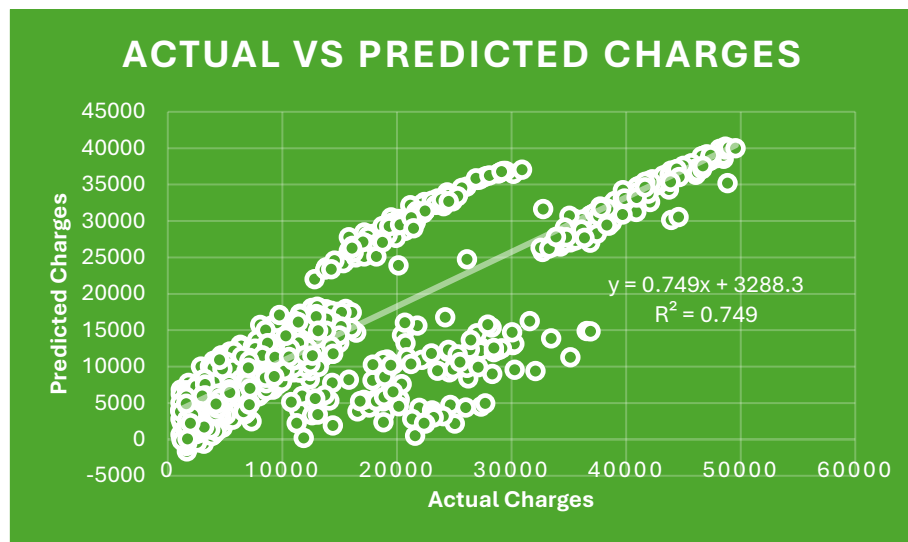
Where:

- Y = Predicted Charges
- β_0 = Intercept (constant term)
- $\beta_1, \beta_2, \beta_3$ = Coefficients for the independent variables (age, BMI, smoker)
- X1, X2, X3 = Values for the independent variables (age, BMI, smoker)

Linear Regression Analysis in Excel:

SUMMARY OUTPUT								
Regression Statistics								
Multiple R	0.865419204							
R Square	0.748950398							
Adjusted R Square	0.74837283							
Standard Error	5888.260843							
Observations	1308							
ANOVA								
	df	SS	MS	F	Significance F			
Regression	3	1.34879E+11	44959754996	1296.730885	0			
Residual	1304	45211786951	34671615.76					
Total	1307	1.80091E+11						
	Coefficients	Standard Error	t Stat	P-value	Lower 95%	Upper 95%	Lower 95.0%	Upper 95.0%
Intercept	-11291.34583	926.68501	-12.18466437	2.0159E-32	-13109.30246	-9473.389192	-13109.30246	-9473.389192
age	257.877189	11.65475117	22.12635733	2.29629E-92	235.0130745	280.7413035	235.0130745	280.7413035
bmi	312.5182505	27.43984691	11.38921261	1.02345E-28	258.6871739	366.349327	258.6871739	366.349327
smoker	23212.6876	405.2450993	57.28061274	0	22417.68389	24007.6913	22417.68389	24007.6913

3.2 Model Performance



This chart is a scatter plot comparing **actual charges** to **predicted charges** from a regression model.

(i) Axes:

- The **X-axis** displays the **actual charges**, which are the true values observed in the dataset.
- The **Y-axis** shows the **predicted charges** which is the values estimated by the model.

(ii) Regression Line:

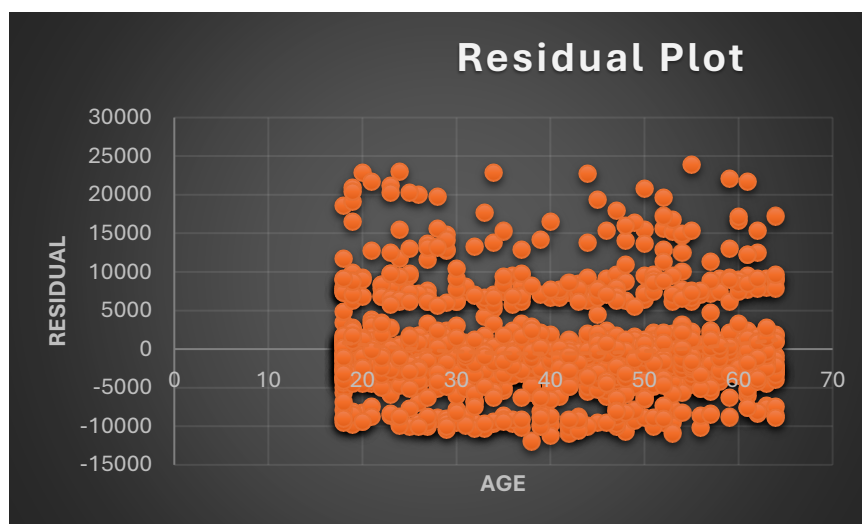
- The equation of this line is given as: $y = 0.749x + 3288.3$. Here, y refers to the predicted charge, 0.749 is the **slope** (indicating that for each unit increase in actual charges, predicted charges rise by 0.749 units), and 3288.3 is the **intercept**, meaning the model predicts a baseline charge of 3288.3 when actual charges are zero.

(iii) R^2 Value (Coefficient of Determination):

- The **R^2 value of 0.749** shows how well the model's predictions match the actual data. An R^2 of 0.749 means that 74.9% of the variation in actual charges is explained by the model, indicating a reasonably good fit but with room for improvement. An R^2 of 1.0 would indicate a perfect fit.

Prediction Result: This chart illustrates that while the model captures the general trend in the data well, the predictions are not perfect. There is some scatter around the line, and the slope being less than 1 suggests the model may underpredict charges for larger values.

3.3 Analysis from Residual Plot: evaluates how well the regression model fits the data by plotting the differences



(i). Axes:

- The **X-axis** represents **Age**, the independent variable in this case.
- The **Y-axis** shows the **residuals**, which indicate the difference between the predicted and actual values:
 - **Positive residuals** (above 0) show that the model **underestimated** the actual value.
 - **Negative residuals** (below 0) mean the model **overestimated** the actual value.

(ii) Distribution of Residuals:

- The residuals are scattered randomly across different ages, suggesting that the model's prediction errors are fairly distributed and not tied to specific age groups.
- The spread of residuals appears relatively even as age increases, which indicates consistent model performance across the age range.

(iii) Potential Issues:

- A few **outliers** with large residuals (both positive and negative) can be seen, particularly around age 20 and 60, indicating that the model significantly mis-predicted charges for certain individuals.
- Although most residuals are near zero, a cluster of positive residuals implies that the model may tend to underpredict charges for certain age groups.

(iv) Summary:

- The plot indicates that the model performs reasonably well across all ages, with no clear bias in residual distribution. However, the presence of outliers and clusters of larger residuals suggests areas where the model could be improved, particularly for certain age groups.

4. Conclusions and Recommendations

4.1 Key Insights

From the analysis, We learned several important things:

- **Smoking** is a major factor that increases medical charges significantly.
- Both **age** and **BMI** are also important predictors of healthcare costs, but the number of children has little effect.
- The model explains a good amount of the variance in medical charges, but there's still room for improvement.

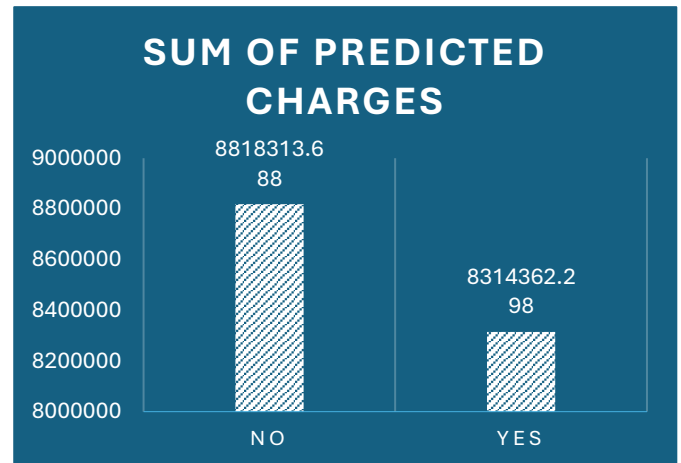
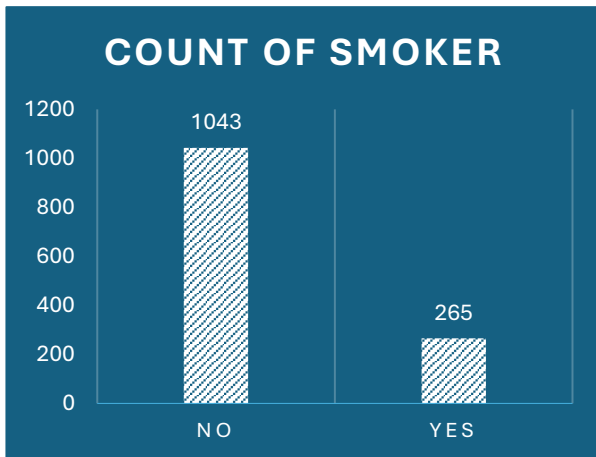
4.2 Recommendations

To enhance the model's predictive accuracy, I suggest:

- **Exploring Additional Variables:** Consider adding more features, like lifestyle factors or pre-existing health conditions, to capture the remaining **25.1%** of the variance.

- **Testing Different Modeling Techniques:** Trying non-linear models or including interaction terms (e.g., between age and smoking status) might improve the model fit.

4.3 Calculated Insights and Conclusion:



- When we consider the number of smokers, There are 265 smokers and 1043 non-smokers.
- When we sum the predicted charges for smokers it is 8,314,362.30 and 31,374.95 on average/person
- Summing up the predicted charges of non-smokers it is around 8,818,313.69 and 8,454.76 on an average/person

With all the above insights, it is clear that, smoker spend much for their medical expenses than the non-smoker according to the model prediction.